Detangling Hydrogel and Microbial Effects of BioBead on Soil Physiochemical Properties and Wheat Yield

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2026-03-06

Introduction

BioBead is a biodegradable hydrogel pellet containing fungal and bacterial consortia for soil health and regenerative agriculture. Microbes in this consortia can fix nitrogen (N), solubilize phosphorus (P), and drive carbon (C) cycling. Arbuscular mycorrhizal fungi (AMF) are obligate mutualists that make direct contact with plant roots and hand off N and P to plants, obtaining C in return. They also provide structures for microbial residence and travel. Greenhouse studies with BioBead have demonstrated improved growth for tomatoes and wheat plants, increases in soil organic matter, and promotion of closed nutrient cycling.

In 2025, a greenhouse study led by Renee Davis and Korena Mafune, PhD evaluated the impacts of 14 different microbe inoculations and treatments on soil microbiology, wheat growth, root colonization, and greenhouse gas emissions. This specific project will focus on soil geochemistry, nutrient dynamics, and soil moisture data, with other data being incorporated in subsequent phases. The responses measured at different intervals will have to be analyzed separately, or combined in another manner. These results will provide a foundation for expansion into larger field trials and BioBead formulation improvements in 2026 and beyond, and hopefully become publications and dissertation chapters.

Research Questions

1. Does the hydrogel carrier in BioBead explain variation in the soil geochemical profile and wheat yield?
2. How does the interaction of the hydrogel and microbes explain variation in the soil geochemical profile and wheat yield?
3. What are the principle components driving variation in these multivariate soil responses?

Methods

Greenhouse Trial and Sample Collection

Wheat seedlings (Red Fife variety, Territorial Seeds) were planted in 1-gallon pots containing sunshine mix and sand (1:2 v/v) from September-December 2025 at the UW Biology Greenhouse. The experiment utilized a randomized design with 14 treatments and 7 biological replicates per treatment (98 pots total). Microbial inoculants were applied at planting according to treatment specifications.

Plants were grown for 14 weeks under greenhouse conditions. Plants were watered weekly with 200 mL water, and soil moisture was recorded 24 hours after each watering event. Leachate samples were collected

at weeks 0, 2, 4, 6, 8 and analyzed using The Gallery (ThermoFisher) for ammonium, nitrate, and phosphate concentrations.

Plant Root Simulator (PRS) Probes (Western Ag Innovations) were buried at week 6 and retrieved after 14 days. Probes were sent to Western Ag for analysis of soil cation and anion exchange capacity via HPLC. At experiment termination (week 14), above-ground biomass was harvested and weighed, and rhizosphere soil and root samples were collected and stored at -80°C for future DNA extraction and metagenomic sequencing. (These data are mentioned for reference and will not be analyzed in this project.)

Data

Files

2025biobeadgreenhouse.Rmd - R Markdown file containing this document.

The data are compiled into a single spreadsheet (**2025greenhousemaster.csv**), which can be found in the GitHub repository. It was compiled from the following raw data.

- **Leachates_2025.csv** - Ammonium, nitrate, and phosphate leachate concentrations recorded throughout the study.
- **PRS_GH_2025.csv** - Soil anion and cation exchange data from PRS probes.
- **WheatMoistureData_2025.csv** - Soil moisture data recorded throughout the study.

N2O_flux.data.csv - Nitrous oxide (N2O) flux was also calculated. The data are limited to 3 replicates per treatment (not 7), so these data will remain in its own file rather than grafted to the master spreadsheet.

Wheat yield data has been added to the master spreadsheet, and includes 2 columns on grain yield (per sample unit) and an associated Harvest Index (HI), which is a ratio of grain yield over the rest of the above-ground biomass. It represents an economically-relevant measure of wheat crop productivity.

Exploratory one-way ANOVAs have already been performed on leachate, soil moisture, anion/cation supply rate, wheat yield, and N2O flux.

Response variables

Code	Full Name	Units	Range
NH4-N_ugml	Ammonium	ug/mL	0.01-5.24
NO3-N_ugml	Nitrate	ug/mL	0-97.55
PO4-P_ugml	Phosphate	ug/mL	0.05-31.97
% soil moisture	Percent Soil Moisture	n/a	0-100
NO3-N	Nitrate	ug/10cm^2/14 days	0.8-8.3
NH4-N	Ammonium	ug/10cm^2/14 days	0-2.9
P	Phosphorus	ug/10cm^2/14 days	6.8-23.7
Ca	Calcium	ug/10cm^2/14 days	751-2220
Mg	Magnesium	ug/10cm^2/14 days	158-477
K	Potassium	ug/10cm^2/14 days	146-305
Fe	Iron	ug/10cm^2/14 days	2.5-21.5
Mn	Manganese	ug/10cm^2/14 days	0.3-1.5
Cu	Copper	ug/10cm^2/14 days	0-1
Zn	Zinc	ug/10cm^2/14 days	1.3-3.8

Code	Full Name	Units	Range
B	Boron	ug/10cm^2/14 days	0-0.6
S	Sulphur	ug/10cm^2/14 days	105-490
Pb	Lead	ug/10cm^2/14 days	0-0.9
Al	Aluminum	ug/10cm^2/14 days	3.4-9.5
Cd	Cadmium	ug/10cm^2/14 days	0-0.2
Na	Sodium	ug/10cm^2/14 days	943-2734

Explanatory variables

Code	Full Name	Units	Range/Values
Fungi	Arbuscular mycorrhizal fungi added	Binary	Present/absent
Bacteria	Bacterial consortia added	Binary	Present/absent
Hydrogel	BioBead hydrogel carrier	Binary	Present/absent
Dose	Inoculum dose	Categorical	Full (100%), Half (50%)
GR24	Synthetic strigolactone addition (5 nM)	Binary	Present/absent
Week	Sampling week	Integer	1-8
Treatment	Treatment code	Categorical	T1-T14 (see below)

Treatment codes

T1 - BioBead – bacteria

T2 - BioBead – AMF

T3 - BioBead - bacteria, AMF

T4 - SL-primed BioBead - bacteria, AMF

T5 - SL-primed BioBead - bacteria

T6 - SL-primed BioBead - AMF

T7 - BioBead w/ reapplication at week 8

T8 - ½ fertilizer ½ BioBead

T9 - Liquid consortia

T10 - Commercial biofertilizer (Hardygro)

T11 - Empty bead

T12 - SL-primed empty bead

T13 - Synthetic fertilizer (positive control – 10% dose, 0.6 g/L, 10mL/pot)

T14 - Untreated

Specific replicates of samples are indicated after a dash in the sample code. For example, certain replicates in the PRS data are designated as T1-1, T1-2, T1-3, etc.

Exploratory NMDS

Nonmetric multidimensional scaling (NMDS) was used to visualize multivariate treatment effects across two datasets: a master dataset comprising nutrient leachate (NH_4-N, NO_3-N, PO_4-P), soil moisture, and wheat yield metrics (grain weight, harvest index); and a soil ion dataset comprising 15 ion supply rates measured by Plant Root Simulator (PRS) probes (NO_3-N, NH_4-N, P, Ca, Mg, K, Fe, Mn, Cu, Zn, B, S, Pb, Al, Cd, Na). Prior to ordination, all variables were relativized by column maxima using the `decostand` function to account for differences in scale across the different response variables. Euclidean distances were used and NMDS was performed in three dimensions for the master dataset and two dimensions for the ion dataset using `metaMDS` with 100 random starts. Ordination stress was evaluated to confirm adequate fit. Treatment groups were visualized using convex hull ellipses via the `ggforce` package in R (version 4.5.2), with treatments colored by inoculant class: core BioBead formulations (blues), strigolactone-primed BioBead formulations (greens), bead-only controls (purples), and reference treatments including synthetic fertilizer, liquid consortia, commercial biofertilizer, and untreated negative control.

Soil Properties NMDS

```
library(vegan)
```

```
## Loading required package: permute
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# load in data
data <- read.csv("2025greenhousemaster_hydrogel.csv", header = TRUE, row.names = 1)
ions <- read.csv("ions/PRS_2025_hydrogel.csv", header = TRUE)

# Subset response and explanatory variables
treatments <- data$Treatment_Category
soil_properties <- data[ , !colnames(data) %in% "Treatment_Category"]
soil_properties_rel <- decostand(soil_properties, method = "max") ## Relativize by column maxima

# create Euclidean distance matrix
data_dist <- vegdist(soil_properties_rel, method = "euclidean")

set.seed(1)
nmds_result <- metaMDS(soil_properties_rel,
                       distance = "euclidean",
                       k = 3,
                       trymax = 100)
```

```
## Run 0 stress 0.05036313
## Run 1 stress 0.05036314
## ... Procrustes: rmse 1.165574e-05  max resid 5.087513e-05
## ... Similar to previous best
## Run 2 stress 0.05036304
## ... New best solution
## ... Procrustes: rmse 3.79069e-05  max resid 0.0001751516
## ... Similar to previous best
## Run 3 stress 0.05036303
## ... New best solution
## ... Procrustes: rmse 2.371307e-05  max resid 0.0001021303
## ... Similar to previous best
## Run 4 stress 0.06211247
## Run 5 stress 0.05036304
## ... Procrustes: rmse 0.0001887209  max resid 0.0008834548
## ... Similar to previous best
## Run 6 stress 0.05036303
## ... Procrustes: rmse 2.122189e-05  max resid 8.56309e-05
## ... Similar to previous best
## Run 7 stress 0.05036308
## ... Procrustes: rmse 0.000206877  max resid 0.0009546713
## ... Similar to previous best
## Run 8 stress 0.0621122
## Run 9 stress 0.05036321
## ... Procrustes: rmse 0.000255649  max resid 0.001195777
## ... Similar to previous best
## Run 10 stress 0.0621137
## Run 11 stress 0.05036306
## ... Procrustes: rmse 3.329164e-05  max resid 0.0001453293
## ... Similar to previous best
## Run 12 stress 0.0621128
## Run 13 stress 0.06211386
## Run 14 stress 0.05036321
## ... Procrustes: rmse 0.0002527943  max resid 0.001191873
## ... Similar to previous best
## Run 15 stress 0.05036319
## ... Procrustes: rmse 0.0002490586  max resid 0.001175898
## ... Similar to previous best
## Run 16 stress 0.05036302
## ... New best solution
## ... Procrustes: rmse 0.0001787761  max resid 0.0008473522
## ... Similar to previous best
## Run 17 stress 0.05036311
## ... Procrustes: rmse 0.0002126575  max resid 0.001013868
## ... Similar to previous best
## Run 18 stress 0.05036314
## ... Procrustes: rmse 5.591481e-05  max resid 0.0002540699
## ... Similar to previous best
## Run 19 stress 0.06211368
## Run 20 stress 0.05036311
## ... Procrustes: rmse 0.0002296839  max resid 0.001095191
## ... Similar to previous best
## *** Best solution repeated 4 times
```

```r
nmds_result$stress
```

```
## [1] 0.05036302
```

```r
#create data frame
cluster_df <- data.frame(
  label = rownames(data),
  cluster = as.factor(data$Treatment_Category)
)
head(cluster_df)
```

```
##   label     cluster
## 1  T1-1 BioBead_bac
## 2  T1-2 BioBead_bac
## 3  T1-3 BioBead_bac
## 4  T1-4 BioBead_bac
## 5  T1-5 BioBead_bac
## 6  T1-6 BioBead_bac
```

```r
nmds_scores <- vegan::scores(nmds_result, tidy = TRUE) |>
  filter(score == "sites") |>
  left_join(cluster_df, by = "label")

# color palette
treatment_colors <- c(
  "BioBead_bac"              = "#084594",  # dark blue
  "BioBead_AMF"              = "#FFFF00",  # yellow
  "BioBead_Mixed"            = "#50C878",  # green

  "Empty_Bead"               = "#969696",  # grey
  "Liquid_Mixed"             = "#964B00",  # brown

  "Fertilizer"               = "#FF0000",  # red
  "Negative"                 = "#000000"   # black (negative control)
)

#plot
library(ggplot2)
p1 <- ggplot(nmds_scores, aes(x = NMDS1, y = NMDS2, colour = cluster)) +
  geom_point(size = 3) +
  scale_color_manual(values = treatment_colors, name = "Treatment Group") +
  stat_ellipse(aes(group = cluster), level = 0.95, linetype = "dashed", linewidth = 0.4) +  # potential
  theme_bw() +
  theme(axis.ticks = element_blank(),
        axis.text = element_blank()) +
  labs(x = "NMDS1", y = "NMDS2",
       caption = "Figure 1: NMDS Ordination of Leachate, Soil Moisture, and Wheat Yield")

p1
```

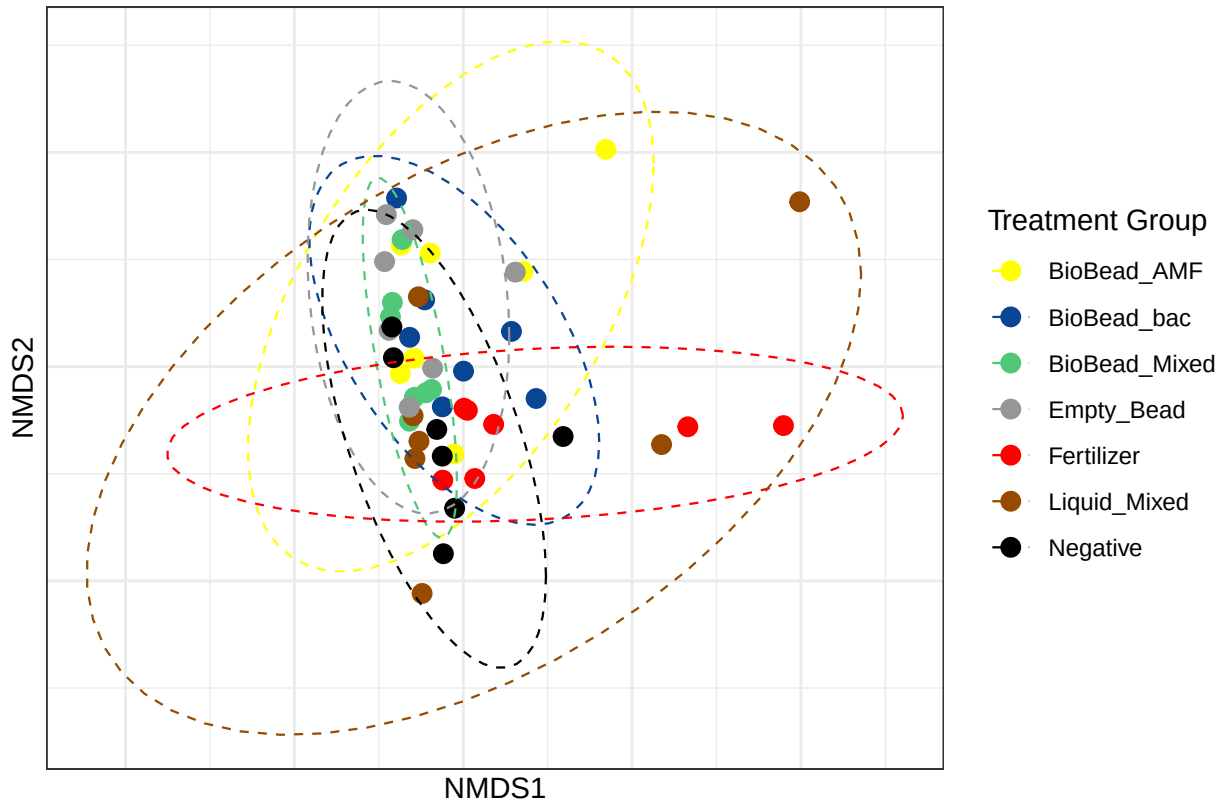


Figure 1: NMDS Ordination of Leachate, Soil Moisture, and Wheat Yield

Based on the ordination, there is no apparent strong multivariate separation between treatments. The fertilizer and liquid consortia (liquid_mixed) is the most distant, indicating that it had unique effects on the response variables. The ellipses are also large, indicating that the variability within the group is high. The BioBead treatments (blue group) have tighter ellipses. This means that they are producing more consistent responses in leachate, soil moisture, and wheat yield.

Soil Ion NMDS

```
library(tidyr)
# Load data
ions <- read.csv("ions/PRS_2025_hydrogel.csv", header = TRUE)

# Subset
ions_treatments <- ions$Treatment_Category
ions_properties <- ions[ , !colnames(ions) %in% c("Notes", "Treatment_Category")]

ions_rel <- decostand(ions_properties, method = "max") # Relativize by column maxima

# NMDS
set.seed(1)
nmds_ions <- metaMDS(ions_rel,
                     distance = "euclidean",
                     k = 3,
                     trymax = 100)
```

```
## Run 0 stress 0.08010406
## Run 1 stress 0.08388827
## Run 2 stress 0.08388828
## Run 3 stress 0.08010469
## ... Procrustes: rmse 0.0009588736  max resid 0.002929448
## ... Similar to previous best
## Run 4 stress 0.08010448
## ... Procrustes: rmse 0.0003963253  max resid 0.0008589719
## ... Similar to previous best
## Run 5 stress 0.0801044
## ... Procrustes: rmse 0.0008042354  max resid 0.002423628
## ... Similar to previous best
## Run 6 stress 0.08388815
## Run 7 stress 0.08388846
## Run 8 stress 0.08010424
## ... Procrustes: rmse 0.0007047422  max resid 0.00215162
## ... Similar to previous best
## Run 9 stress 0.08010397
## ... New best solution
## ... Procrustes: rmse 0.00044346  max resid 0.001391222
## ... Similar to previous best
## Run 10 stress 0.08388828
## Run 11 stress 0.08010452
## ... Procrustes: rmse 0.0004322618  max resid 0.00130887
## ... Similar to previous best
## Run 12 stress 0.08010484
## ... Procrustes: rmse 0.0005433678  max resid 0.001552972
## ... Similar to previous best
## Run 13 stress 0.08010394
## ... New best solution
## ... Procrustes: rmse 8.769406e-05  max resid 0.0002668914
## ... Similar to previous best
## Run 14 stress 0.08010446
## ... Procrustes: rmse 0.0006782236  max resid 0.001736295
## ... Similar to previous best
## Run 15 stress 0.08010455
## ... Procrustes: rmse 0.0005081741  max resid 0.001495404
## ... Similar to previous best
## Run 16 stress 0.08010399
## ... Procrustes: rmse 0.0001214049  max resid 0.0003746663
## ... Similar to previous best
## Run 17 stress 0.08388838
## Run 18 stress 0.08010465
## ... Procrustes: rmse 0.0005744307  max resid 0.001683421
## ... Similar to previous best
## Run 19 stress 0.0801044
## ... Procrustes: rmse 0.0004418141  max resid 0.001330737
## ... Similar to previous best
## Run 20 stress 0.08010411
## ... Procrustes: rmse 0.0004757364  max resid 0.001181081
## ... Similar to previous best
## *** Best solution repeated 7 times
```

```r
nmds_ions$stress
```

```
## [1] 0.08010394
```

```r
#create data frame
cluster_df_ions <- data.frame(
  label = rownames(ions),
  cluster = as.factor(ions$Treatment_Category)
)
head(cluster_df_ions)
```

```
##   label     cluster
## 1     1 BioBead_bac
## 2     2 BioBead_bac
## 3     3 BioBead_bac
## 4     4 BioBead_AMF
## 5     5 BioBead_AMF
## 6     6 BioBead_AMF
```

```r
nmds_scores_ions <- vegan::scores(nmds_ions, tidy = TRUE) |>
  filter(score == "sites") |>
  left_join(cluster_df_ions, by = "label")

library(ggforce)
p2 <- ggplot(nmds_scores_ions, aes(x = NMDS1, y = NMDS2, colour = cluster)) +
  geom_point(size = 3) +
  scale_color_manual(values = treatment_colors, name = "Treatment Group") +
  stat_ellipse(aes(group = cluster), level = 0.95, linetype = "dashed", linewidth = 0.4) +
  theme_bw() +
  theme(axis.ticks = element_blank(),
        axis.text = element_blank()) +
  labs(x = "NMDS1", y = "NMDS2",
       caption = "Figure 2: NMDS Ordination of Soil Ion Supply Rates by Treatment")
p2
```

```
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_path()`).
```

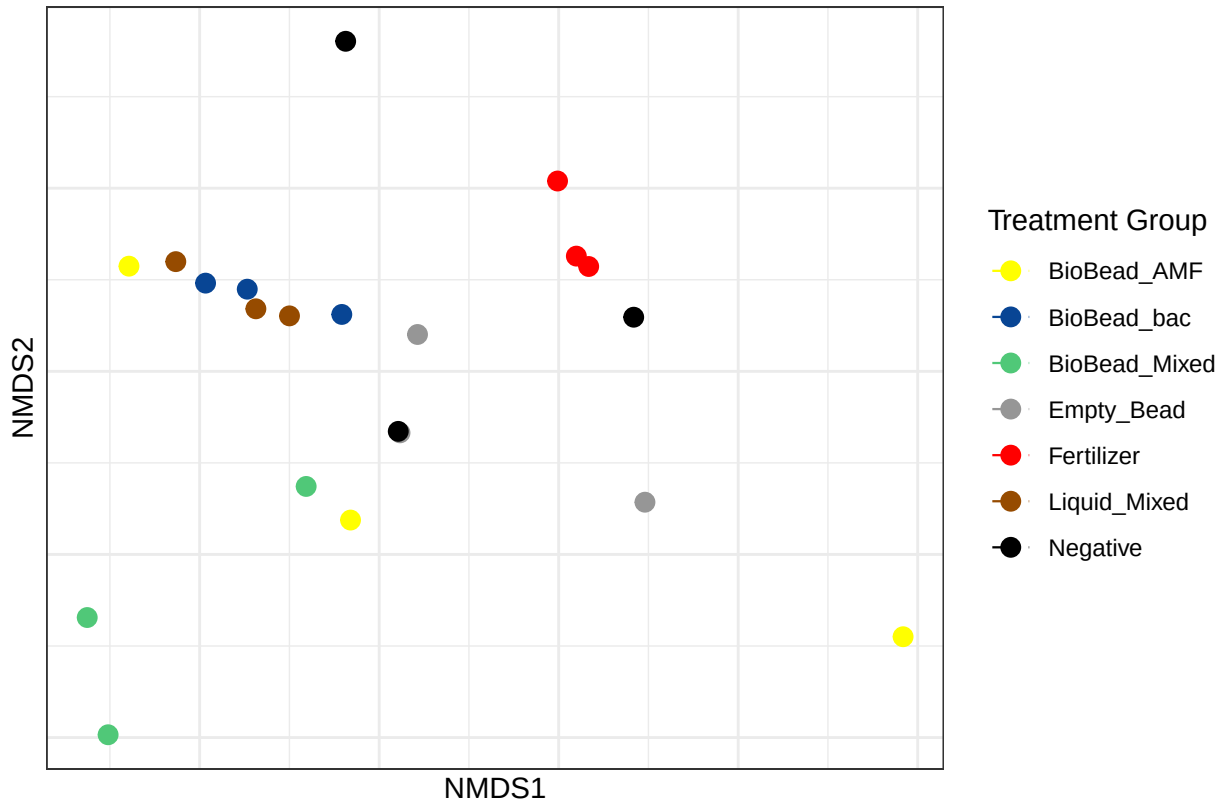


Figure 2: NMDS Ordination of Soil Ion Supply Rates by Treatment

```
p1 <- ggplot(nmds_scores, aes(x = NMDS1, y = NMDS2, colour = cluster)) +
  geom_point(size = 3) +
  scale_color_manual(values = treatment_colors, name = "Treatment Group") +
  stat_ellipse(aes(group = cluster), level = 0.95, linetype = "dashed", linewidth = 0.4) +  # potential
  theme_bw() +
  theme(axis.ticks = element_blank(),
        axis.text = element_blank()) +
  labs(x = "NMDS1", y = "NMDS2",
       caption = "Figure 1: NMDS Ordination of Leachate, Soil Moisture, and Wheat Yield")

p1
```

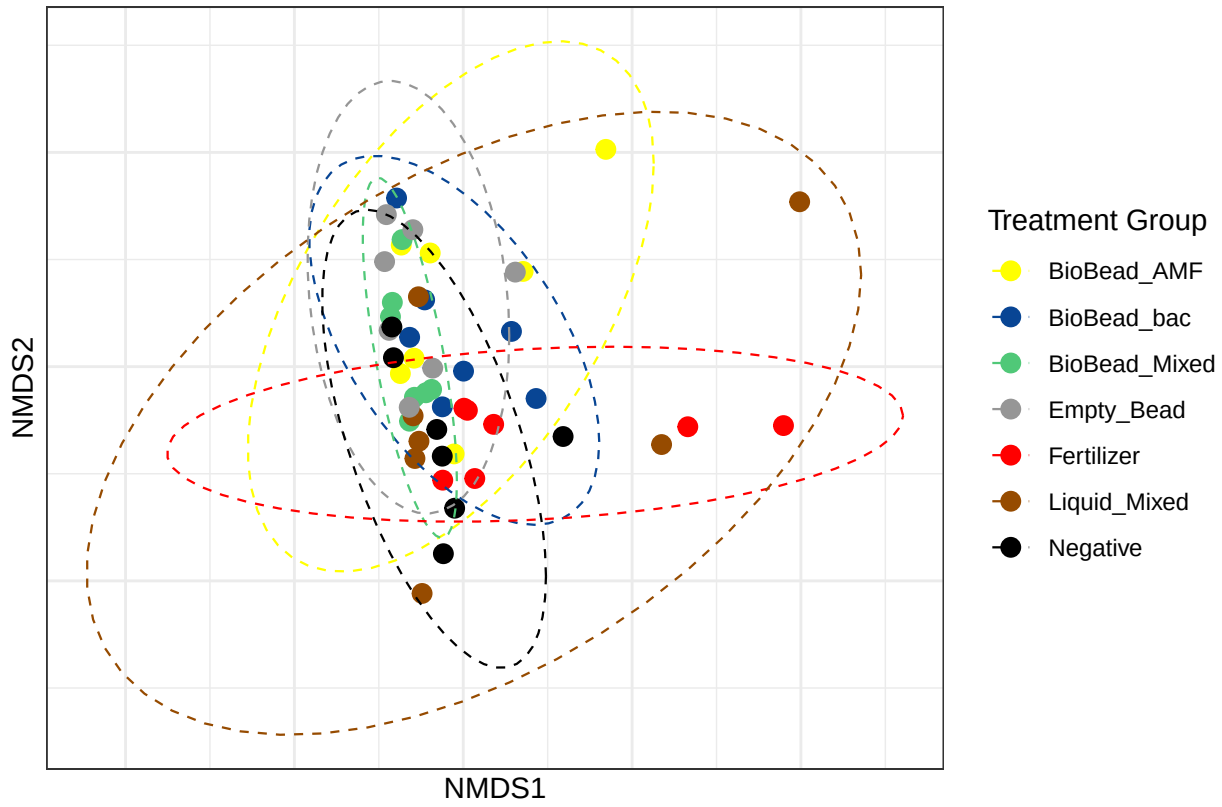


Figure 1: NMDS Ordination of Leachate, Soil Moisture, and Wheat Yield

The ion ordination was also a generally good fit for the original data (stress 0.111). It shows a slightly different story than the previous ordination. Here, the BioBead treatments have larger ellipses, highlighting more within-group variability. Fertilizer and liquid_mixed are tighter. The strigolactone (SL) primed treatments are pulling to the upper left of the ordination space, and are clustering together. This generally shows that soil ions have different patterning than the other 3 response variables in regards to treatment categories.

Hydrogel influences on soil physiochemical properties

Soil physiochemical properties

The analysis of hydrogel influences will take place in 2 parts: the soil physiochemical properties (leachate, moisture, wheat yield) and soil ions.

PERMANOVA

First, I will conduct a PERMANOVA on a select number of treatment categories:

1. T3 - BioBead - bacteria, AMF
2. T9 - Liquid consortia
3. T11 - Empty bead

The hypothesis is that there are differences in soil physiochemical properties for these different treatment groups. To do this, I will filter the dataset and make a new distance matrix. Permutations not restricted, as the study utilized a randomized block design so there are no bench effects to account for.

```r
# Filter to 3 treatments
soil_sub <- soil_properties_rel[treatments %in% c("BioBead_Mixed", "Liquid_Mixed", "Empty_Bead"), ]
treatments_sub <- treatments[treatments %in% c("BioBead_Mixed", "Liquid_Mixed", "Empty_Bead")]

# Recompute distance matrix on subset
dist_sub <- vegdist(soil_sub, method = "euclidean")

# PERMANOVA
set.seed(1)
gel_permanova_result <- adonis2(dist_sub ~ treatments_sub,
                                permutations = 9999)
gel_permanova_result
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = dist_sub ~ treatments_sub, permutations = 9999)
##          Df SumOfSqs      R2     F Pr(>F)
## Model     2   1.7218 0.32766 4.386  1e-04 ***
## Residual 18   3.5331 0.67234
## Total    20   5.2549 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this test, the treatment group explained 32.8% of the multivariate variance ($R^2 = 0.328$), with a highly significant F statistic ($F = 4.386$, $p = 0.0001$). This confirms the hypothesis that there are differences in soil physiochemical properties in these different treatment groups.

Practically, these data mean that the 3 treatment groups produce meaningfully different multivariate soil/yield profiles. These three treatments isolate the carrier and microbial effects, so the separation suggests the hydrogel carrier and microbial consortia are each contributing something distinct rather than being interchangeable.

Two-Factor PERMANOVA

I can use the approach of a two-factor PERMANOVA to understand how much variance is captured by the presence or absence of the hydrogel, the presence or absence of the microbes, and the interaction between the two. In order to do this, I manually created 2 additional columns in the data files in Excel to indicate hydrogel Y/N and microbe Y/N. Treatments were coded with hydrogels and microbes even if present in small amounts (e.g. BioBead_Mixed_Reduced, HalfF_HalfBB). Hardygro was marked as present for microbe.

```r
data_gel2factor <- read.csv("2025greenhousemaster_gel2factor.csv", header = TRUE, row.names = 1)

# Subset
soil_properties <- data_gel2factor[ , !colnames(data_gel2factor) %in% c("Treatment_Category", "Hydrogel

# Relativize/redo distance matrix
soil_properties_rel <- decostand(soil_properties, method = "max")
gel2factor_dist <- vegdist(soil_properties_rel, method = "euclidean")

set.seed(1)
```

```r
gel_permanova_2way <- adonis2(
  gel2factor_dist ~ Hydrogel * Microbe,
  data = data_gel2factor,
  permutations = 9999,
  by = "term"
)
gel_permanova_2way
```

```
## Permutation test for adonis under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = gel2factor_dist ~ Hydrogel * Microbe, data = data_gel2factor, permutations = 9999,
##                   Df SumOfSqs      R2      F Pr(>F)
## Hydrogel           1   1.6503 0.13891 8.4106 0.0001 ***
## Microbe            1   1.1662 0.09817 5.9437 0.0001 ***
## Hydrogel:Microbe   1   0.2338 0.01968 1.1914 0.3069
## Residual          45   8.8296 0.74324
## Total             48  11.8799 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The hydrogel accounts for 13.9% of the variance in the responses ($F = 8.4106$, $p = 0.0001$). The presence or absence of microbe accounts for 9.8% of the variance and is also highly significant ($F = 5.9437$, $p = 0.0001$). Together, their interaction accounts for 2% of the variance but is not statistically significant ($F = 1.1914$, $p = 0.3069$). The R2 from Hydrogel, Microbe, and their interaction is 0.25676, meaning that these factors can explain 25.7% of the multivariate response.

RDA

Following this PERMANOVA test, a redundancy analysis (RDA) can help visualize what treatments drive separation of responses.

```r
gel_rda <- rda(
  soil_properties_rel ~ Hydrogel * Microbe,
  data = data_gel2factor,
  scale = FALSE
)

summary(gel_rda)
```

```
##
## Call:
## rda(formula = soil_properties_rel ~ Hydrogel * Microbe, data = data_gel2factor,      scale = FALSE)
##
## Partitioning of variance:
##               Inertia Proportion
## Total         0.24750     1.0000
## Constrained   0.06355     0.2568
## Unconstrained 0.18395     0.7432
```

```
##
## Eigenvalues, and their contribution to the variance
##
## Importance of components:
##                            RDA1    RDA2     RDA3     PC1     PC2     PC3     PC4
## Eigenvalue            0.03486 0.02428 0.004414 0.08127 0.04381 0.03011 0.01908
## Proportion Explained  0.14083 0.09809 0.017836 0.32837 0.17703 0.12164 0.07708
## Cumulative Proportion 0.14083 0.23892 0.256760 0.58513 0.76215 0.88379 0.96087
##                             PC5      PC6
## Eigenvalue            0.006354 0.003331
## Proportion Explained  0.025671 0.013457
## Cumulative Proportion 0.986543 1.000000
##
## Accumulated constrained eigenvalues
## Importance of components:
##                            RDA1    RDA2     RDA3
## Eigenvalue            0.03486 0.02428 0.004414
## Proportion Explained  0.54851 0.38203 0.069465
## Cumulative Proportion 0.54851 0.93053 1.000000
```

```
# Significance test of the full model
anova(gel_rda, permutations = 9999)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 9999
##
## Model: rda(formula = soil_properties_rel ~ Hydrogel * Microbe, data = data_gel2factor, scale = FALSE)
##          Df Variance      F Pr(>F)
## Model     3 0.063548 5.1819  1e-04 ***
## Residual 45 0.183951
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(gel_rda, by = "term", permutations = 9999)
```

```
## Permutation test for rda under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 9999
##
## Model: rda(formula = soil_properties_rel ~ Hydrogel * Microbe, data = data_gel2factor, scale = FALSE)
##                  Df Variance      F Pr(>F)
## Hydrogel          1 0.034381 8.4106 0.0001 ***
## Microbe           1 0.024297 5.9437 0.0001 ***
## Hydrogel:Microbe  1 0.004870 1.1914 0.3109
## Residual         45 0.183951
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Plot
plot(gel_rda, type = "n")
```

```r
points(gel_rda, display = "sites",
       col = treatment_colors[data_gel2factor$Treatment_Category],
       pch = 16, cex = 1.2)
text(gel_rda, display = "bp", col = "red")  # arrows for explanatory variables
legend("topright", legend = names(treatment_colors),
       col = treatment_colors, pch = 16, cex = 0.7)
mtext("Figure 3: RDA of Soil and Yield Properties constrained by Hydrogel and Microbe presence",
      side = 1, # bottom of plot
      line = 4, # how far below the axis
      cex = 0.8) # font size
```

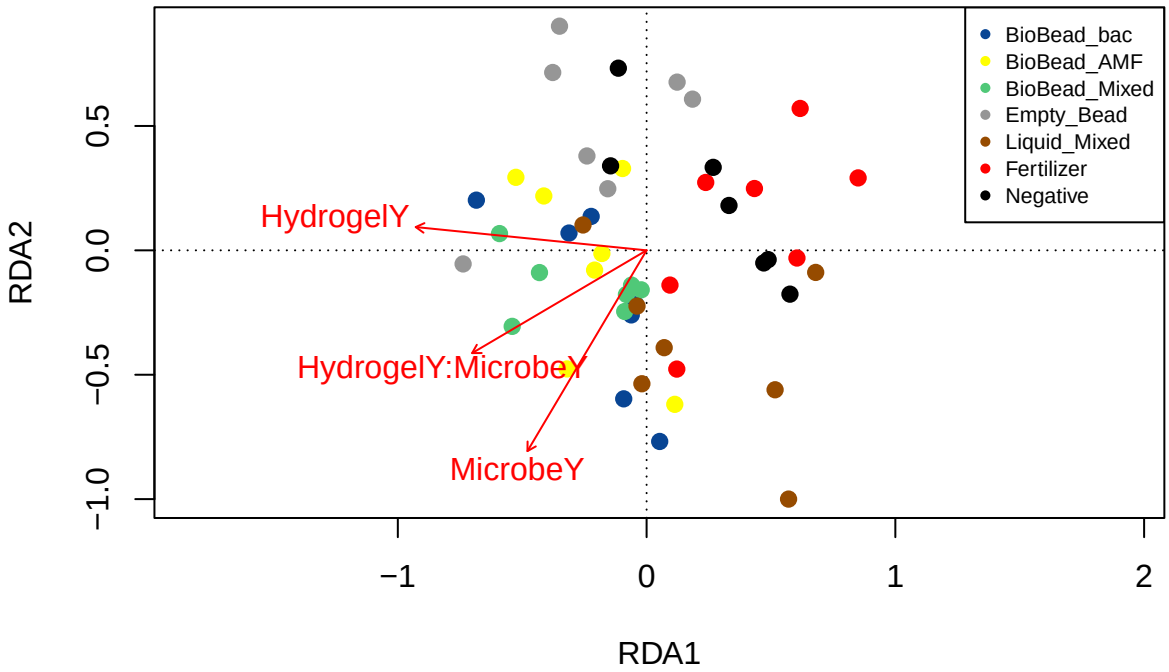


Figure 3: RDA of Soil and Yield Properties constrained by Hydrogel and Microbe presence

From this RDA, we can see a few trends. The Hydrogel (Y) arrow points to the upper left, and so does the interaction arrow (Hydrogel Y/Microbe Y). This means that the interaction is largely due to the hydrogel signal. The BioBead treatments (blue dots) are also in that direction. The Microbe (Y) arrow points in a different direction, so it's separate from the hydrogel effect. The Fertilizer treatments group together, and is also in the opposite direction to MicrobeY.

In examining the partitioning of the variance, these these factors (hydrogel and microbe) are driving 23.9% of the multivariate response. This is consistent with the PERMANOVA R2 above. Therefore, Hydrogel and Microbe presence are meaningful predictors of soil physiochemical properties (leachate, moisture, and wheat yield), but there are other drivers for the remaining 75% of these multivariate responses.

PCA

```
library(tidyverse)
```

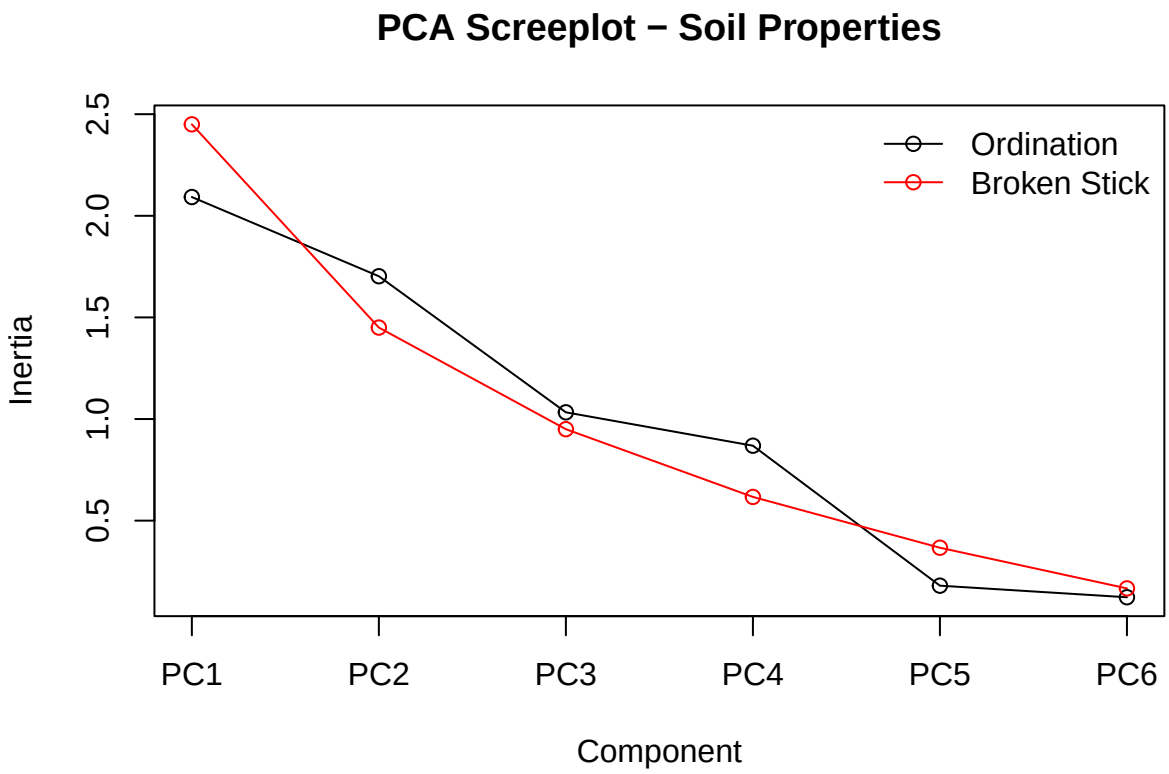
```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.1.5
## v lubridate 1.9.4     v stringr   1.6.0
## v purrr     1.2.0     v tibble    3.3.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Uses soil_properties_rel already built above
# scale = TRUE equalizes variance across variables (z-score equivalent)
soil_pca <- rda(soil_properties_rel, scale = TRUE)

# How much variance does each PC explain?
summary(soil_pca)
```

```
##
## Call:
## rda(X = soil_properties_rel, scale = TRUE)
##
## Partitioning of correlations:
##               Inertia Proportion
## Total               6          1
## Unconstrained       6          1
##
## Eigenvalues, and their contribution to the correlations
##
## Importance of components:
##                            PC1    PC2    PC3    PC4    PC5     PC6
## Eigenvalue              2.0924 1.7027 1.0331 0.8687 0.1800 0.12310
## Proportion Explained    0.3487 0.2838 0.1722 0.1448 0.0300 0.02052
## Cumulative Proportion 0.3487 0.6325 0.8047 0.9495 0.9795 1.00000
```

```
# Scree plot - where does variance flatten out?
screeplot(soil_pca, bstick = TRUE, type = "lines",
          main = "PCA Screeplot - Soil Properties")
```




```
# Extract proportion of variance for axis labels
pca_summary <- summary(soil_pca)
pc1_var <- round(pca_summary$cont$importance[2,1] * 100, 1)  # % variance PC1
pc2_var <- round(pca_summary$cont$importance[2,2] * 100, 1)  # % variance PC2

# Build scores data frame (same pattern as NMDS)
pca_scores <- vegan::scores(soil_pca, tidy = TRUE) |>
  filter(score == "sites") |>
  left_join(cluster_df, by = "label")

# Extract species (variable) scores for arrows
pca_arrows <- vegan::scores(soil_pca, display = "species") |>
  as.data.frame() |>
  rownames_to_column("variable")

# Plot
p_pca <- ggplot(pca_scores, aes(x = PC1, y = PC2, colour = cluster)) +
  geom_point(size = 3) +
  scale_color_manual(values = treatment_colors, name = "Treatment Group") +
  # Add variable arrows
  geom_segment(data = pca_arrows,
               aes(x = 0, y = 0, xend = PC1, yend = PC2),
               arrow = arrow(length = unit(0.2, "cm")),
               color = "darkgray", inherit.aes = FALSE) +
  geom_text(data = pca_arrows,
            aes(x = PC1 * 1.15, y = PC2 * 1.15, label = variable),
```

```
            color = "darkgray", size = 3, inherit.aes = FALSE) +
  stat_ellipse(aes(group = cluster), level = 0.95,
               linetype = "dashed", linewidth = 0.4) +
  theme_bw() +
  theme(axis.ticks = element_blank(),
        axis.text = element_blank()) +
  labs(x = paste0("PC1 (", pc1_var, "%)"),
       y = paste0("PC2 (", pc2_var, "%)"),
       caption = "Figure 5: PCA of Soil Physiochemical Properties and Wheat Yield")

p_pca
```

```
## Warning in MASS::cov.trob(data[, vars], wt = weight * nrow(data)): Probable
## convergence failure
```

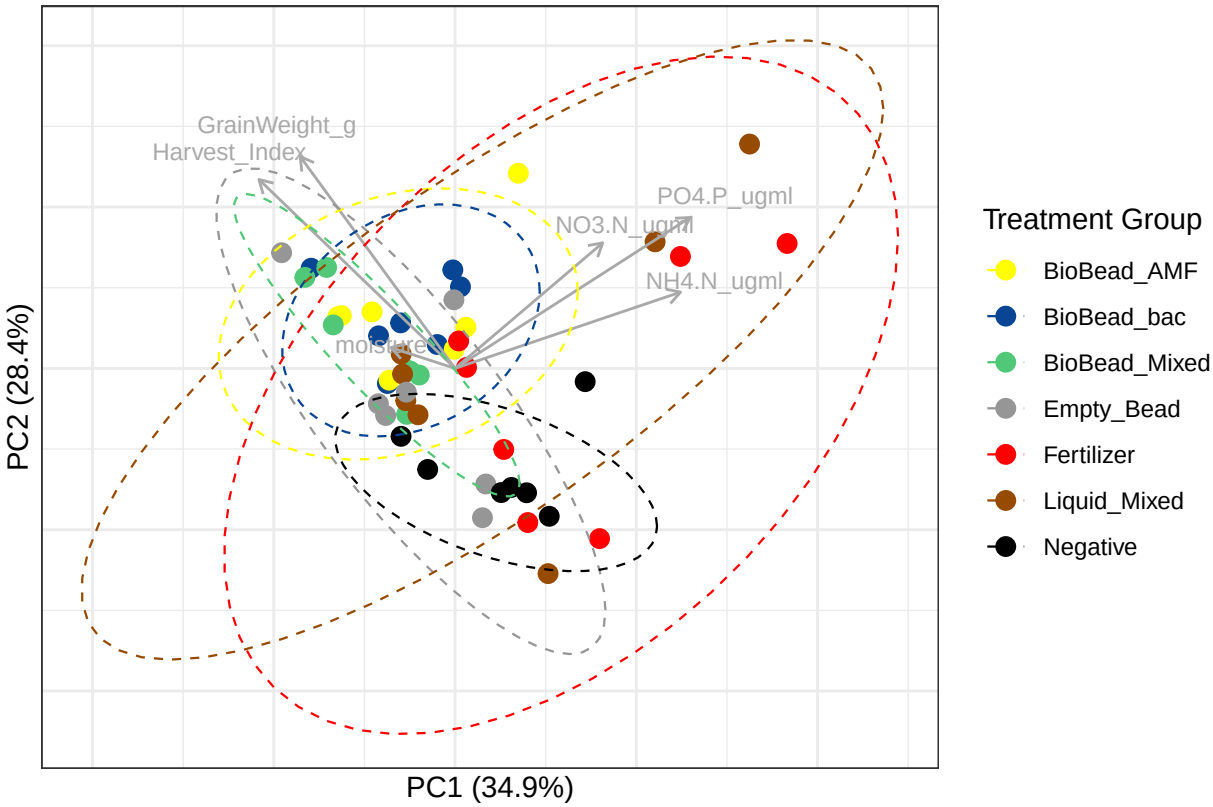


Figure 5: PCA of Soil Physiochemical Properties and Wheat Yield

Soil ions

PERMANOVA

I will use the same PERMANOVA approach using these treatment categories:

1. T3 - BioBead - bacteria, AMF
2. T9 - Liquid consortia

3. T11 - Empty bead

The hypothesis is that there are differences in soil ion supply rates for these different treatment groups. To do this, I will filter the dataset and make a new distance matrix. Same as above, the permutations not restricted, as the study utilized a randomized block design so there are no bench effects to account for.

```
# Filter to 3 treatments
ions_treatments <- ions$Treatment_Category
ion_sub <- ions_rel[ions_treatments %in% c("BioBead_Mixed", "Liquid_Mixed", "Empty_Bead"), ]
ion_treatments_sub <- ions_treatments[ions_treatments %in% c("BioBead_Mixed", "Liquid_Mixed", "Empty_Bea

# Recompute distance matrix on subset
ion_dist_sub <- vegdist(ion_sub, method = "euclidean")

# PERMANOVA
set.seed(1)
ion_permanova_result <- adonis2(ion_dist_sub ~ ion_treatments_sub,
                                permutations = 9999)
ion_permanova_result
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = ion_dist_sub ~ ion_treatments_sub, permutations = 9999)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     2   3.6525 0.59729 4.4496 0.0037 **
## Residual  6   2.4626 0.40271
## Total     8   6.1151 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this test, the treatment group explained 59.7% of the multivariate variance ($R^2 = 0.59729$), with a highly significant F statistic ($F = 4.4496$, $p = 0.0037$). This confirms the hypothesis that there are differences in soil ion supply rate in these different treatment groups.

Practically, these data mean that the 3 treatment groups produce meaningfully different multivariate ion supply rates. Like in the soil physiochemical profile above, these three treatments isolate the carrier and microbial effects, so the separation suggests the hydrogel carrier and microbial consortia are each contributing something distinct rather than being interchangeable.

Two-Factor PERMANOVA

Using the same data modification approach in the previous section, I coded each treatment in the ion dataset with a Hydrogel Y/N and a Microbe Y/N for a Two-Factor ANOVA.

```
ion_gel2factor <- read.csv("ions/PRS_2025_gel2factor.csv", header = TRUE, row.names = 1)

# Subset
ion_properties <- ion_gel2factor[ , !colnames(ion_gel2factor) %in% c("Treatment_Category", "Hydrogel",

# Relativize/redo distance matrix
ion_properties_rel <- decostand(ion_properties, method = "max")
```

```r
ion2factor_dist <- vegdist(ion_properties_rel, method = "euclidean")

set.seed(1)
ion_permanova_2way <- adonis2(
  ion2factor_dist ~ Hydrogel * Microbe,
  data = ion_gel2factor,
  permutations = 9999,
  by = "term"
)
ion_permanova_2way
```

```
## Permutation test for adonis under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = ion2factor_dist ~ Hydrogel * Microbe, data = ion_gel2factor, permutations = 9999, b
##                  Df SumOfSqs      R2      F Pr(>F)
## Hydrogel          1   1.2224 0.07614 1.8652 0.0908 .
## Microbe           1   2.3076 0.14374 3.5212 0.0070 **
## Hydrogel:Microbe  1   1.3829 0.08614 2.1102 0.0613 .
## Residual         17  11.1411 0.69397
## Total            20  16.0540 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Similar to the other multivariate responses in the previous section, the hydrogel accounts for only 6.7% of the variance in the responses ($F = 1.8652$, $p = 0.0908$). But this result is not statistically significant. The presence or absence of microbe accounts for 14.4% of the variance and is very significant ($F = 3.5212$, $p = 0.0070$). Together, their interaction accounts for 8.6% of the variance and is borderline statistically significant ($F = 2.1102$, $p = 0.0613$). So the presence of the hydrogel is not significant in terms of its impacts on the soil ion supply rate. It is worth noting that there are fewer sample units compared to soil physiochemical properties (42 instead of 98), so the lack of significance may be a result of a comparatively under-powered test.

RDA

```r
ion_rda <- rda(
  ion_properties_rel ~ Hydrogel * Microbe,
  data = ion_gel2factor,
  scale = FALSE
)
summary(ion_rda)
```

```
##
## Call:
## rda(formula = ion_properties_rel ~ Hydrogel * Microbe, data = ion_gel2factor,      scale = FALSE)
##
## Partitioning of variance:
##               Inertia Proportion
```

```
## Total          0.8027      1.000
## Constrained    0.2456      0.306
## Unconstrained  0.5571      0.694
##
## Eigenvalues, and their contribution to the variance
##
## Importance of components:
##                           RDA1    RDA2    RDA3    PC1    PC2     PC3     PC4
## Eigenvalue            0.1371 0.06909 0.03942 0.2350 0.1077 0.06823 0.04331
## Proportion Explained  0.1709 0.08607 0.04910 0.2928 0.1342 0.08500 0.05396
## Cumulative Proportion 0.1709 0.25692 0.30603 0.5988 0.7330 0.81800 0.87196
##                             PC5     PC6     PC7     PC8     PC9     PC10     PC11
## Eigenvalue            0.02712 0.02572 0.01679 0.01388 0.01095 0.004631 0.001488
## Proportion Explained  0.03379 0.03204 0.02092 0.01729 0.01364 0.005769 0.001854
## Cumulative Proportion 0.90575 0.93779 0.95871 0.97599 0.98964 0.995405 0.997259
##                            PC12      PC13      PC14      PC15      PC16
## Eigenvalue            0.001066 0.0007159 0.0002241 0.0001395 5.553e-05
## Proportion Explained  0.001327 0.0008919 0.0002792 0.0001737 6.917e-05
## Cumulative Proportion 0.998586 0.9994779 0.9997571 0.9999308 1.000e+00
##
## Accumulated constrained eigenvalues
## Importance of components:
##                           RDA1    RDA2    RDA3
## Eigenvalue            0.1371 0.06909 0.03942
## Proportion Explained  0.5583 0.28124 0.16046
## Cumulative Proportion 0.5583 0.83954 1.00000
```

```
# Significance test of the full model
set.seed(1)
anova(ion_rda, permutations = 9999)
```

```
## Permutation test for rda under reduced model
## Permutation: free
## Number of permutations: 9999
##
## Model: rda(formula = ion_properties_rel ~ Hydrogel * Microbe, data = ion_gel2factor, scale = FALSE)
##          Df Variance      F Pr(>F)
## Model     3  0.24565 2.4989 0.0041 **
## Residual 17  0.55705
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(ion_rda, by = "term", permutations = 9999)
```

```
## Permutation test for rda under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 9999
##
## Model: rda(formula = ion_properties_rel ~ Hydrogel * Microbe, data = ion_gel2factor, scale = FALSE)
##                  Df Variance      F Pr(>F)
## Hydrogel          1  0.06112 1.8652 0.0905 .
## Microbe           1  0.11538 3.5212 0.0069 **
```

```
## Hydrogel:Microbe  1  0.06915 2.1102 0.0640 .
## Residual         17  0.55705
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Plot (run together)
plot(ion_rda, type = "n")
points(ion_rda, display = "sites",
       col = treatment_colors[ion_gel2factor$Treatment_Category],
       pch = 16, cex = 1.2)
text(ion_rda, display = "bp", col = "red")  # arrows for explanatory variables
legend("topright", legend = names(treatment_colors),
       col = treatment_colors, pch = 16, cex = 0.7)
mtext("Figure 4: RDA of Soil Ion Supply Rate constrained by Hydrogel and Microbe presence",
      side = 1, # bottom of plot
      line = 4, # how far below the axis
      cex = 0.8) # font size
```

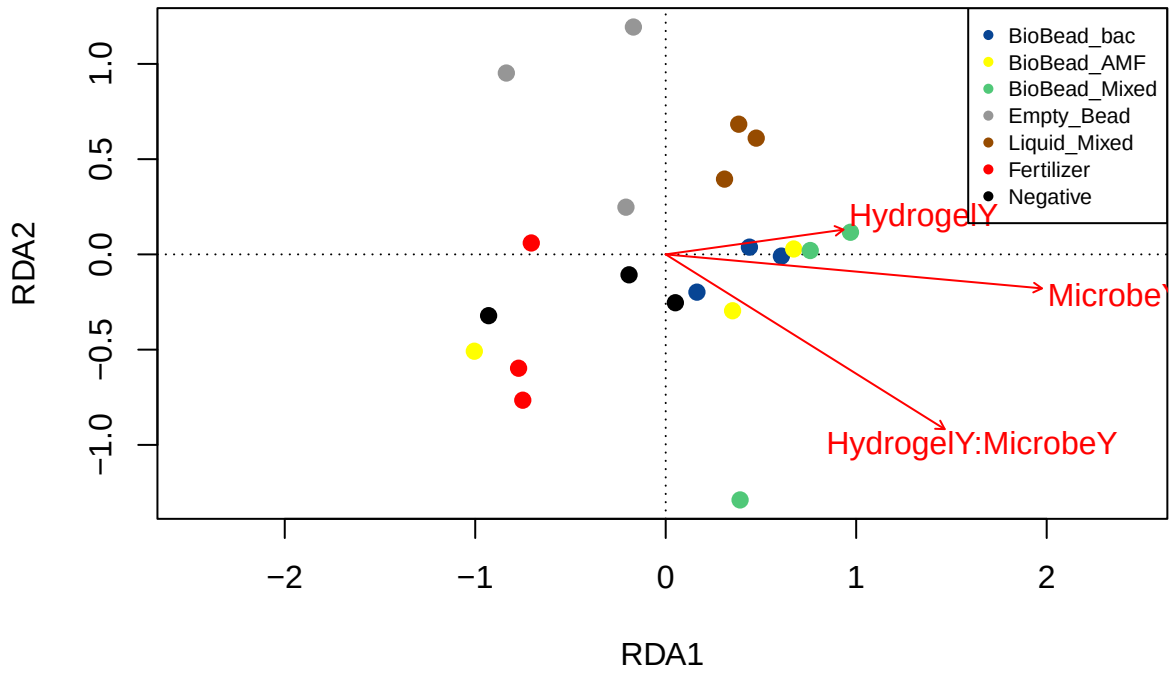


Figure 4: RDA of Soil Ion Supply Rate constrained by Hydrogel and Microbe presence

In examining the breakdowns of the RDA axes, RDA1 explains 9.7% of the variation (F = 2.1324, p = 0.0062) and there is a nice separation of treatments along this axis. RDA2 picks up another 3.6%, and RDA3 another 1.2%. There is still significant variation found in the unconstrained axes (~86%) that is not explained by Hydrogel or Microbe factors.

The arrows for Hydrogel and Microbe are pointing in similar directions, meaning that these explanatory variables are correlated with each other and driving similar patterns in the ion supply rate responses. It is worth noting that treatments that have microbes but no hydrogel are few in this data set (Liquid_Mixed,

HardyGro). As a result of this design, it may be difficult to complelety detangle these factors. It could be worthwhile to remove some lower priority treatment categories like HalfFert_HalfBB, BB_Mixed_Reduced to balance the analysis for a cleaner interpretation.

The proportion of constrained variance is 0.1441 or 14.41%, meaning that Hydrogel + Microbe and their interactions account for 14% of the variation in response of this system. This is higher than 10.7% with the leachate, moisture, and wheat yield responses in the previous section. So these factors are having a stronger influence on soil ion dynamics than these other response variables. It makes biological sense that microbial activity and hydrogel water retention would have more direct effects on ion exchange at the root zone than on other metrics.

The ion data in general may be telling a cleaner story. The PRS probes are gathering ion exchange near the rhizosphere over a 2-week period. It is closer to the nutrient exchange occuring between microbes and plant roots than leachate, moisture, and wheat yield.

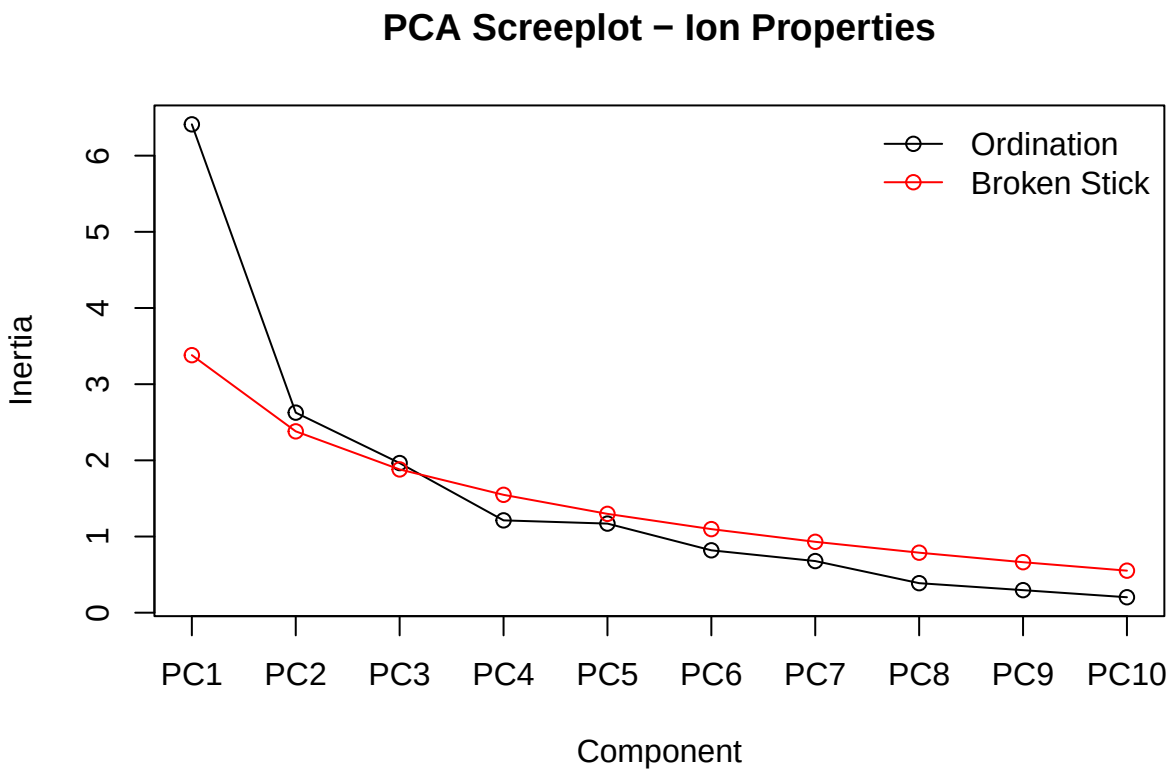
PCA

```
# Uses soil_properties_rel already built above
# scale = TRUE equalizes variance across variables (z-score equivalent)
ion_pca <- rda(ion_properties_rel, scale = TRUE)

# How much variance does each PC explain?
summary(ion_pca)
```

```
##
## Call:
## rda(X = ion_properties_rel, scale = TRUE)
##
## Partitioning of correlations:
##               Inertia Proportion
## Total              16          1
## Unconstrained      16          1
##
## Eigenvalues, and their contribution to the correlations
##
## Importance of components:
##                            PC1    PC2    PC3     PC4    PC5     PC6     PC7
## Eigenvalue            6.4110 2.6263 1.9642 1.21202 1.1696 0.81879 0.67777
## Proportion Explained  0.4007 0.1641 0.1228 0.07575 0.0731 0.05117 0.04236
## Cumulative Proportion 0.4007 0.5648 0.6876 0.76334 0.8364 0.88762 0.92998
##                            PC8    PC9    PC10     PC11     PC12     PC13
## Eigenvalue            0.38760 0.2960 0.20289 0.113302 0.061085 0.023537
## Proportion Explained  0.02422 0.0185 0.01268 0.007081 0.003818 0.001471
## Cumulative Proportion 0.95420 0.9727 0.98538 0.992465 0.996283 0.997754
##                           PC14     PC15      PC16
## Eigenvalue            0.018056 0.011984 0.0059001
## Proportion Explained  0.001129 0.000749 0.0003688
## Cumulative Proportion 0.998882 0.999631 1.0000000
```

```
# Scree plot - where does variance flatten out?
screeplot(ion_pca, bstick = TRUE, type = "lines",
          main = "PCA Screeplot - Ion Properties")
```




```
# Extract proportion of variance for axis labels
ionpca_summary <- summary(ion_pca)
ionpc1_var <- round(ionpca_summary$cont$importance[2,1] * 100, 1)  # % variance PC1
ionpc2_var <- round(ionpca_summary$cont$importance[2,2] * 100, 1)  # % variance PC2

cluster_df_ions <- data.frame(
  label = rownames(ion_properties_rel),
  cluster = as.factor(ion_gel2factor$Treatment_Category)
)

# Build scores data frame (same pattern as NMDS)
ionpca_scores <- vegan::scores(ion_pca, tidy = TRUE) |>
  filter(score == "sites") |>
  left_join(cluster_df_ions, by = "label")

# Extract species (variable) scores for arrows
ionpca_arrows <- vegan::scores(ion_pca, display = "species") |>
  as.data.frame() |>
  rownames_to_column("variable")

# Plot
ionp_pca <- ggplot(ionpca_scores, aes(x = PC1, y = PC2, colour = cluster)) +
  geom_point(size = 3) +
  scale_color_manual(values = treatment_colors, name = "Treatment Group") +
  # Add variable arrows
  geom_segment(data = ionpca_arrows,
```

```
              aes(x = 0, y = 0, xend = PC1, yend = PC2),
              arrow = arrow(length = unit(0.2, "cm")),
              color = "darkgray", inherit.aes = FALSE) +
  geom_text(data = ionpca_arrows,
            aes(x = PC1 * 1.15, y = PC2 * 1.15, label = variable),
            color = "darkgray", size = 3, inherit.aes = FALSE) +
  stat_ellipse(aes(group = cluster), level = 0.95,
               linetype = "dashed", linewidth = 0.4) +
  theme_bw() +
  theme(axis.ticks = element_blank(),
        axis.text = element_blank()) +
  labs(x = paste0("PC1 (", ionpc1_var, "%)"),
       y = paste0("PC2 (", ionpc2_var, "%)"),
       caption = "Figure 6: PCA of Soil Ion Supply Rate")

ionp_pca
```

```
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
## Too few points to calculate an ellipse
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_path()`).
```

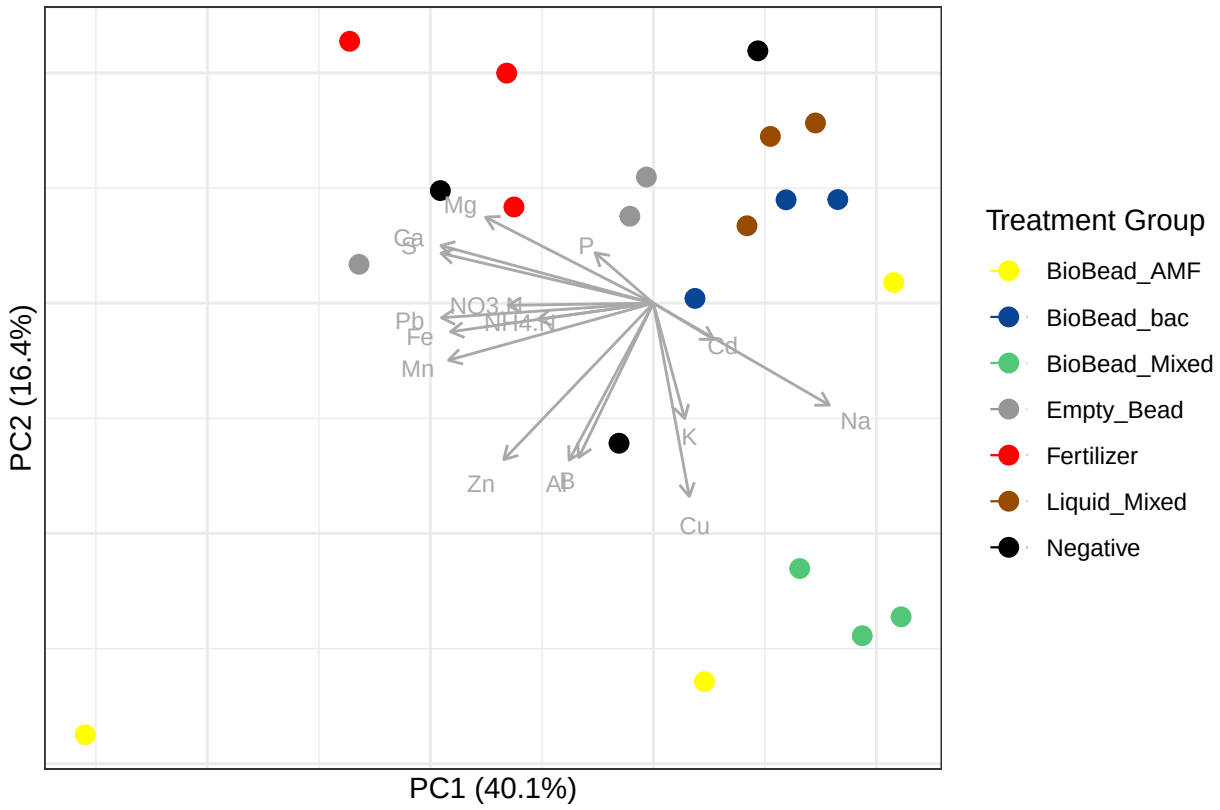


Figure 6: PCA of Soil Ion Supply Rate

References

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Jariani P, Sabokdast M, Rajabi F, Naghavi MR, Dedicova B. Enhancing salinity tolerance in wheat: the role of synthetic Strigolactone (GR24) in modulating antioxidant enzyme activities, ion channels, and related gene expression in stress responses. Sci Rep. 2025 Jul 7;15(1):24216. doi: 10.1038/s41598-025-08670-3. PMID: 40624098; PMCID: PMC12234850.

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Sedaghat, M., Emam, Y., Mokhtassi-Bidgoli, A., Hazrati, S., Lovisolo, C., Visentin, I., Cardinale, F., & Tahmasebi-Sarvestani, Z. (2021). The Potential of the Synthetic Strigolactone Analogue GR24 for the Maintenance of Photosynthesis and Yield in Winter Wheat under Drought: Investigations on the Mechanisms of Action and Delivery Modes. Plants, 10(6), 1223. https://doi.org/10.3390/plants10061223